CCNA Preparation – Chapter 1: Network Devices

◆ What are Network Devices?

Network devices are hardware or software components used to connect computers and other devices, enabling communication, data transfer, and network management within and between networks.

◆ Types of Network Devices

1. 🔌 Switch

Definition:
A switch is a networking device that connects multiple devices within the same Local Area Network (LAN) and forwards data based on MAC addresses.

Key Points:

- Has many ports (usually 24+)
- Connects end devices like PCs, printers
- Works within a single LAN
- Does NOT connect different networks

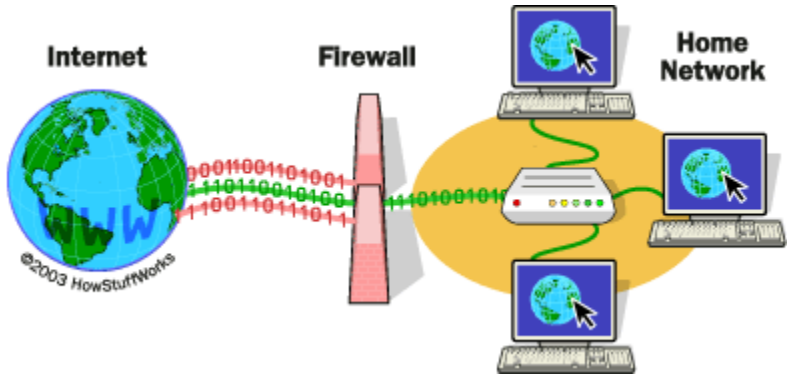
2. 🌐 Router

Definition:
A router is a networking device that connects multiple networks and routes data between them using IP addresses.

Key Points:

- Connects different LANs
- Used to access the Internet
- Uses IP addresses for routing
- Has fewer interfaces than switches

3. 🔥 Firewall



Definition:
A firewall is a security device that monitors and controls incoming and outgoing network traffic based on predefined rules.

Key Points:

- Protects networks from unauthorized access
- Can be hardware or software
- Filters traffic between networks
- Types:
  - Network Firewall (hardware-based)
  - Host-based Firewall (software-based)

◆ Summary

- **Switch** → Connects devices within a LAN
- **Router** → Connects different networks
- **Firewall** → Secures the network

◆ Real-World Example

- In a company:
  - Switch connects employees' computers
  - Router connects office to the Internet
  - Firewall protects against cyber attacks